CONTENT

ABSTRACT

Printed Circuit Boards (PCBs) are the backbone of modern electronic devices, providing the foundation for electrical connectivity and functionality. Even minor defects in a PCB can lead to system malfunctions, reduced performance, or complete failure of the device. Traditional manual inspection methods are labor-intensive, time-consuming, and prone to human error, making them impractical for large-scale industrial production where precision and speed are critical. To overcome these challenges, this project presents the design and implementation of a computer vision–based application for automatic detection and classification of PCB defects.

The proposed system leverages a combination of image processing techniques and machine learning algorithms to detect and categorize common defects, including missing holes, open circuits, shorts, spurious copper, mouse bites, and missing components. A dataset of PCB images is preprocessed and used for model training, enabling the system to recognize defects with high accuracy. Image subtraction and contour detection methods are employed for localization, while a trained classification model ensures reliable categorization of defect types.

The application further incorporates a user-friendly interface that allows operators to upload PCB images, visualize the detected defects with bounding boxes, and generate a defect summary report in real-time. Experimental results demonstrate significant improvements in inspection speed and accuracy compared to manual and traditional inspection techniques. The system not only reduces inspection costs but also enhances overall quality assurance in PCB manufacturing.

By combining automation, scalability, and precision, the proposed application represents a cost-effective alternative to conventional Automated Optical Inspection (AOI) systems. With further optimization and integration into production pipelines, this solution has the potential to significantly advance modern electronics manufacturing and ensure higher reliability of electronic products.

INTRODUCTION

A Printed Circuit Board (PCB) defect detection and classification system is an application specifically designed for the electronics manufacturing industry to efficiently inspect and ensure the quality of circuit boards. The primary objective behind the development of the “PCB Defect Detection and Classification Application” is to automate the inspection process of PCBs, which traditionally relies on manual checking or costly equipment. Tailored for industries of all scales, the system focuses on identifying common PCB defects that can lead to failures in electronic devices.

The project involves capturing PCB images, preprocessing them, and applying computer vision techniques to detect defects such as missing holes, shorts, open circuits, spurious copper, mouse bites, and missing components. A trained machine learning model is then used to classify these defects accurately. By assigning labels and highlighting the defect regions, the system provides a clear and systematic way to analyze PCB quality.

Given the increasing demand for electronics in every sector, there is a pressing need for a reliable and cost-effective inspection method that can handle large volumes of PCB production while maintaining high accuracy. This system addresses that demand by providing a faster, automated, and more accurate solution compared to traditional manual inspection.

The implementation of this application is expected to enhance the efficiency of PCB manufacturing, reduce human error, and minimize production costs. It empowers engineers and quality control teams to seamlessly identify defects and ensure that only error-free boards move forward in the production line. The overarching goal of this project is to digitize and automate all aspects of PCB defect detection and classification to achieve improved product quality and reliability.

SCOPE OF THE SYSTEM

The PCB Defect Detection and Classification Application is designed to address the operational requirements of electronics manufacturing industries, ranging from small-scale units to large-scale factories producing high volumes of Printed Circuit Boards (PCBs). Its primary goal is to ensure product reliability by optimizing the process of PCB quality inspection and minimizing errors in defect identification.

In contrast to traditional inspection methods, such as manual checking or expensive Automated Optical Inspection (AOI) systems, several challenges arise. Manual inspection is time-consuming, labor-intensive, and prone to human error, while AOI systems involve high setup and maintenance costs that may not be feasible for smaller industries. These limitations create risks of undetected defects, reduced production efficiency, and increased manufacturing costs.

By transitioning to an automated defect detection system powered by computer vision and machine learning, these challenges are effectively addressed. The digital approach enhances inspection accuracy, reduces dependency on human operators, and significantly improves scalability for mass production. Furthermore, the system provides a cost-effective solution compared to high-end industrial machines, making it suitable for industries of all sizes.

Ultimately, the scope of the system extends to improving quality assurance in PCB manufacturing by delivering fast, reliable, and automated defect detection. It enhances overall productivity, ensures higher product reliability, and reduces wastage—thereby supporting the growing demands of the electronics industry.

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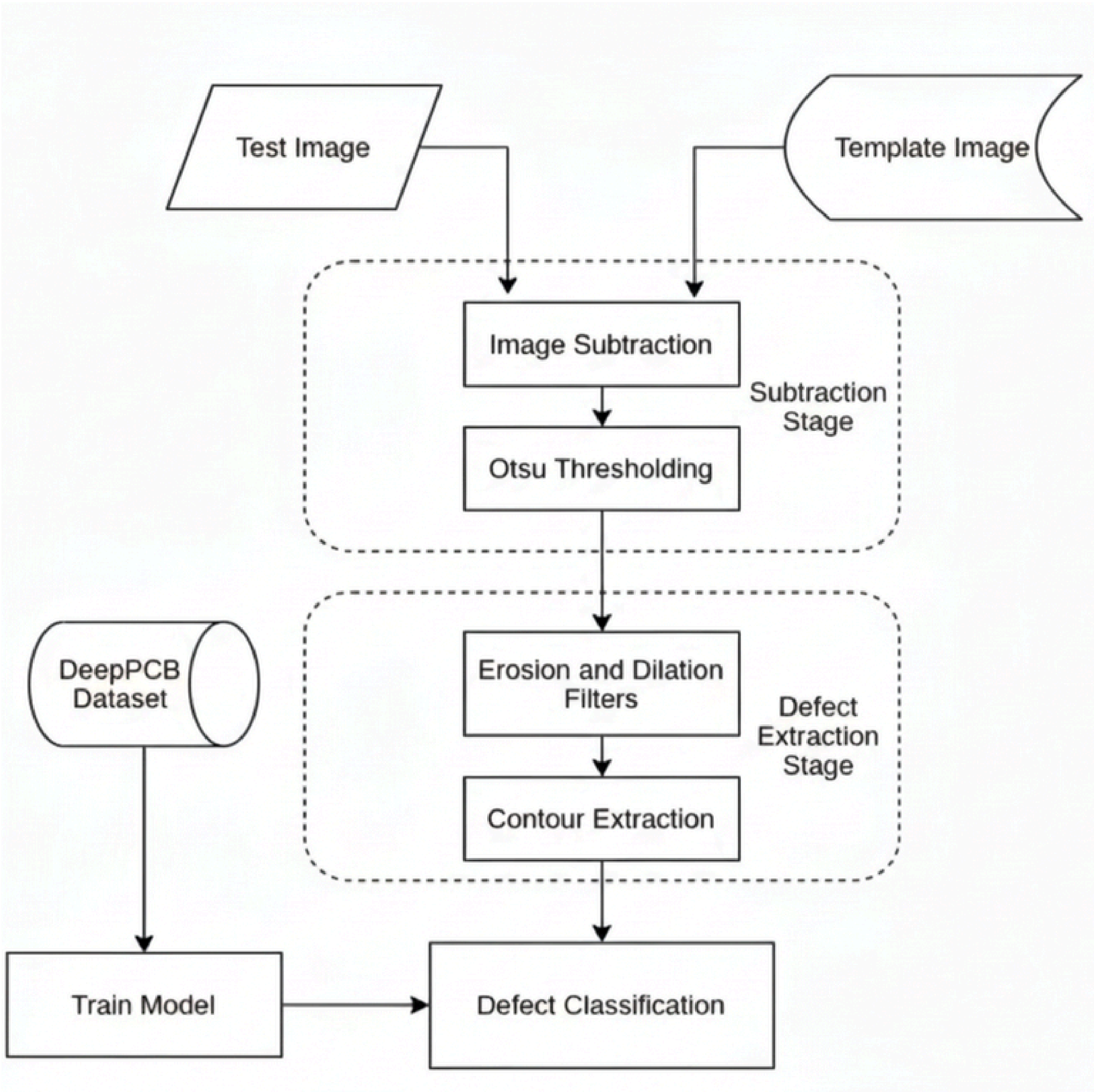
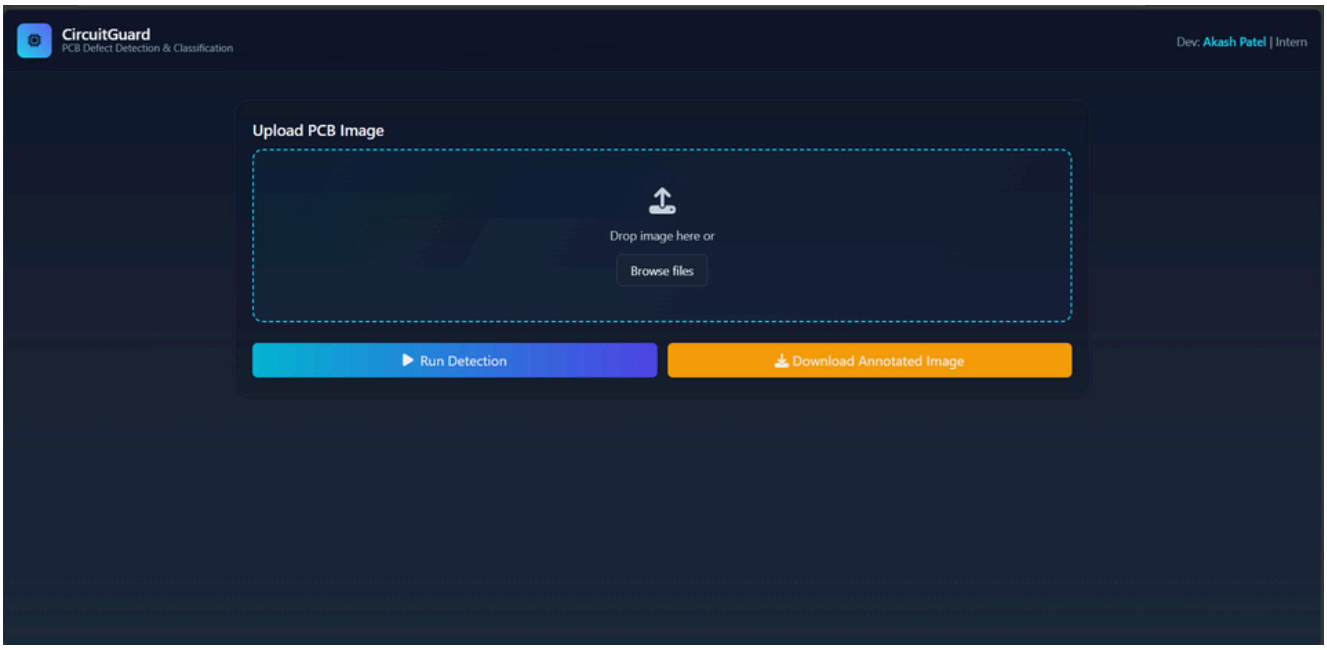
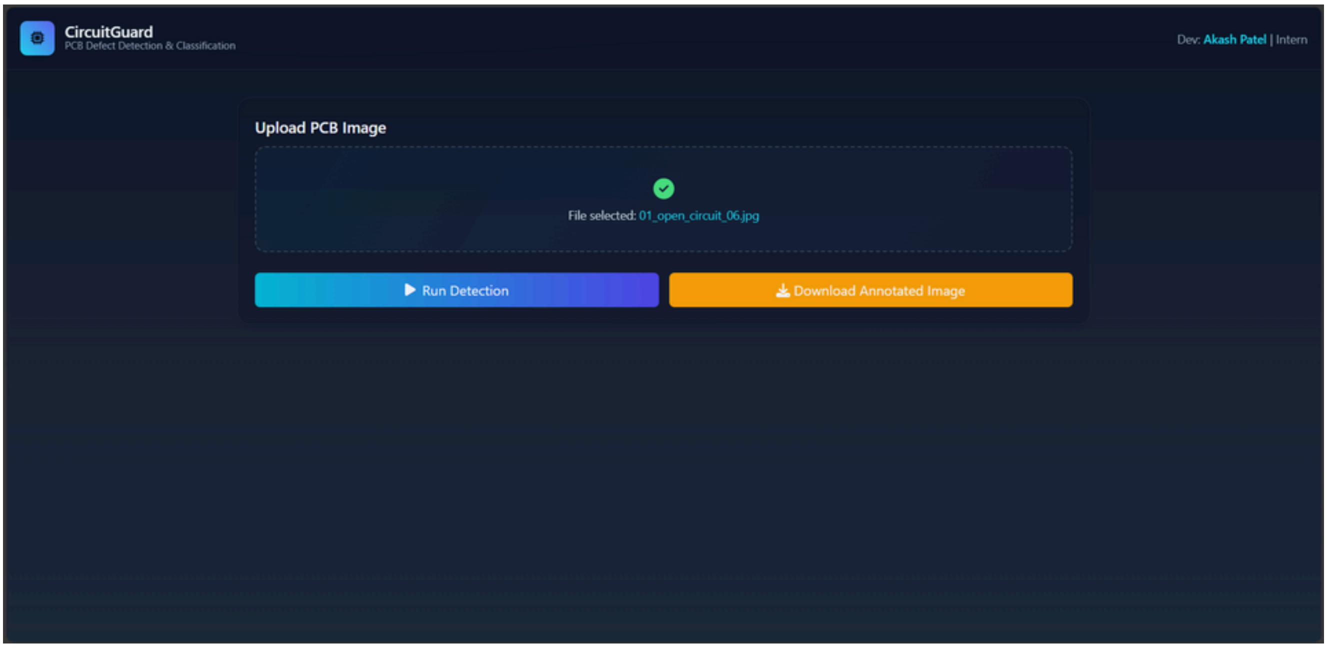
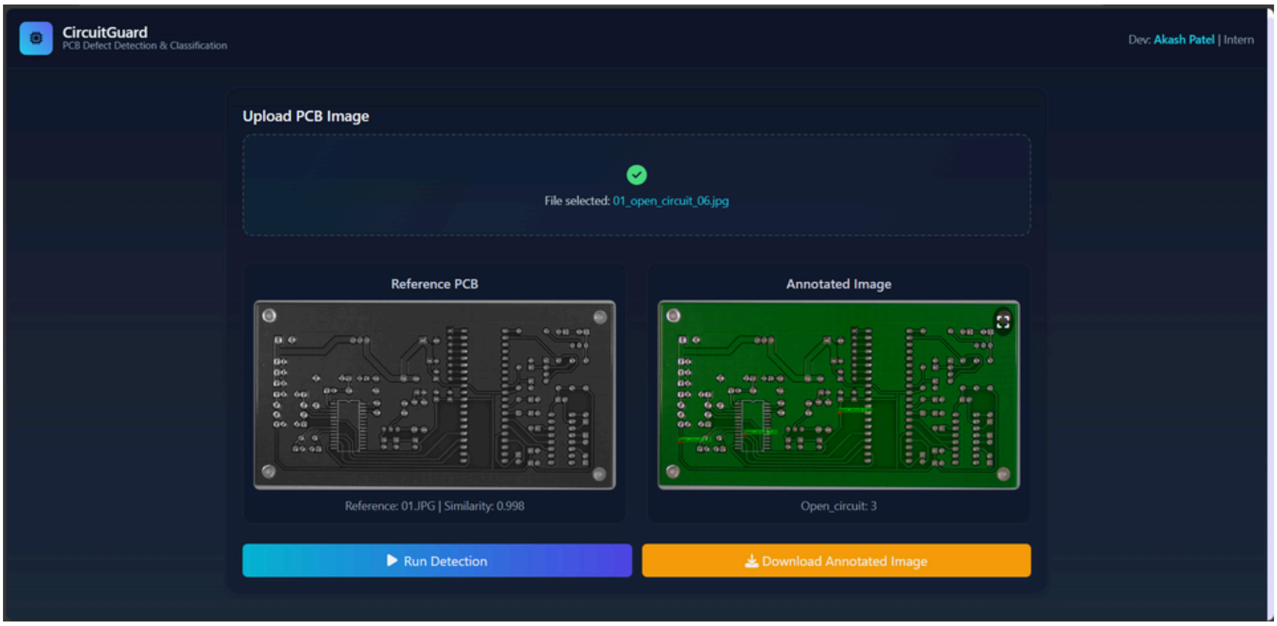


CircuitGuard-PCB Defect Detection and Classification Application

Application Interface Overview






CircuitGuard
PCB Defect Detection & Classification
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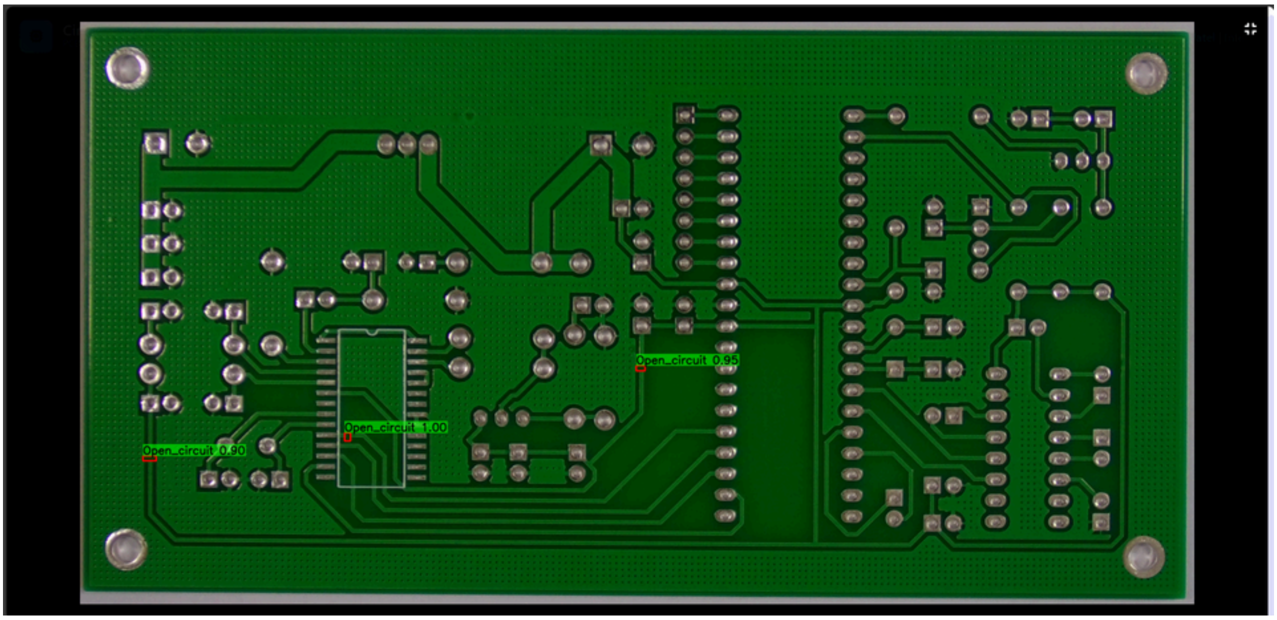
Download Annotated Image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METHODOLOGY & PROGRAMMING

METHODOLOGY :

1. IMAGE ACQUISITION AND PREPROCESSING

- PCB images (template/golden PCB and test PCB) are collected from the DeepPCB dataset.
- Preprocessing techniques such as resizing, grayscale conversion, filtering, and alignment are applied.
- Image subtraction is performed between template and test images to generate a defect difference map.
- Thresholding (Otsu's method) and noise reduction filters highlight possible defect regions.

2. DFECT LOCALIZATION

- OpenCV contour detection is used to locate defect regions.
- Bounding boxes are drawn, and Regions of Interest (ROIs) are extracted.
- Each defect ROI is labeled and stored for training.

3. DEFECT CLASSIFICATION

- Extracted ROIs are classified using a deep learning model.
- EfficientNet-B4 (implemented in PyTorch) is trained with defect datasets (augmented to improve generalization).
- The model outputs categories such as missing hole, short, open circuit, mouse bite, spurious copper, missing component, etc.
- Performance is measured using accuracy, precision, recall, F1-score, and confusion matrices.

4. FRONTEND AND BACKEND INTEGRATION

- A web-based interface (Streamlit/HTML, CSS, JS) allows users to upload PCB images.
- The backend pipeline processes the image, applies the trained model, and returns annotated outputs.
- Results are displayed with bounding boxes and defect labels, along with an option to export reports (annotated image + CSV log).

5. TESTING AND OPTIMIZATION

- The system is tested on unseen PCB images to evaluate performance.
- Pipeline is optimized for speed (≤ 5 seconds per image) and reliability.
- Final validation ensures smooth functioning and industry applicability.

PROGRAMMING

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The programming aspect of the project is modular, covering image processing, model training and UI development.

1. **ANGUAGES AND LIBRARIES USED**
    - Python for core implementation.
    - OpenCV, NumPy for image processing.
    - PyTorch, timm for deep learning (EfficientNet model).
    - Matplotlib, Seaborn for evaluation plots.
    - Streamlit / HTML, CSS, JS for frontend UI.
2. **KEY MODULES IMPLEMENTED**

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IMAGE PEPROSSING AND SUBSTRACTION:

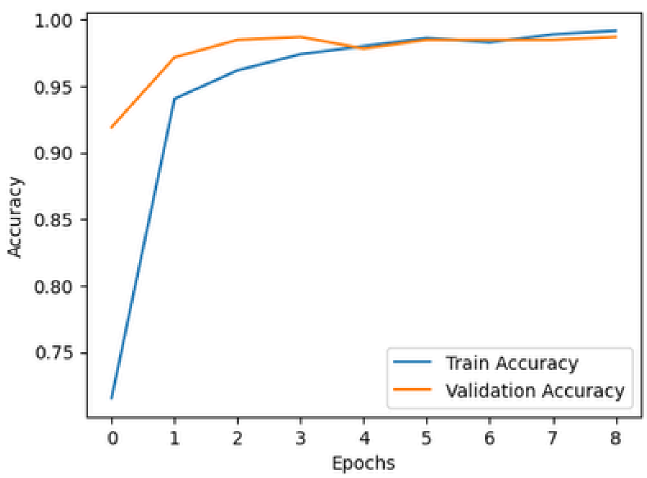
RESULTS & DISCUSSION

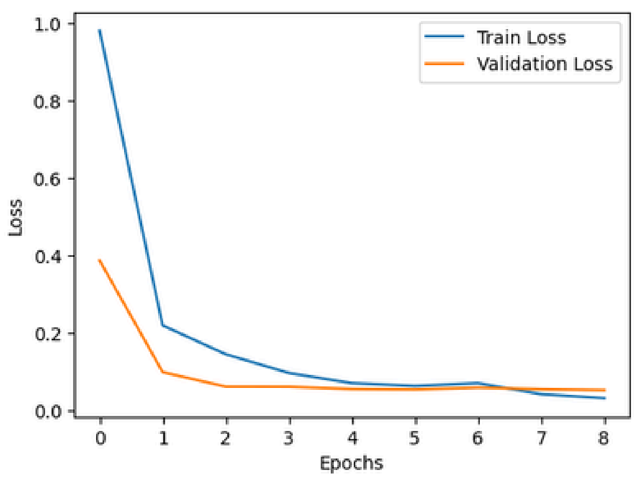
The PCB Defect Detection and Classification Application was successfully developed using computer vision and deep learning techniques. The system was trained and tested using the DeepPCB dataset, which includes multiple categories of PCB defects such as Missing Hole, Mouse Bite, Open Circuit, Short, Spur, and Spurious Copper. The main objective was to automate the defect detection process and achieve high classification accuracy using an optimized deep learning model.

MODEL TRAINING AND PERFORMANCE :

The EfficientNet-B4 architecture was implemented in PyTorch for defect classification due to its high performance and efficient parameter utilization. The dataset was divided into **80% training and 20% validation sets** . The model was trained for **9 epochs** using the Adam optimizer and Cross-Entropy loss function.

The learning curves below illustrate the training and validation performance of the model:



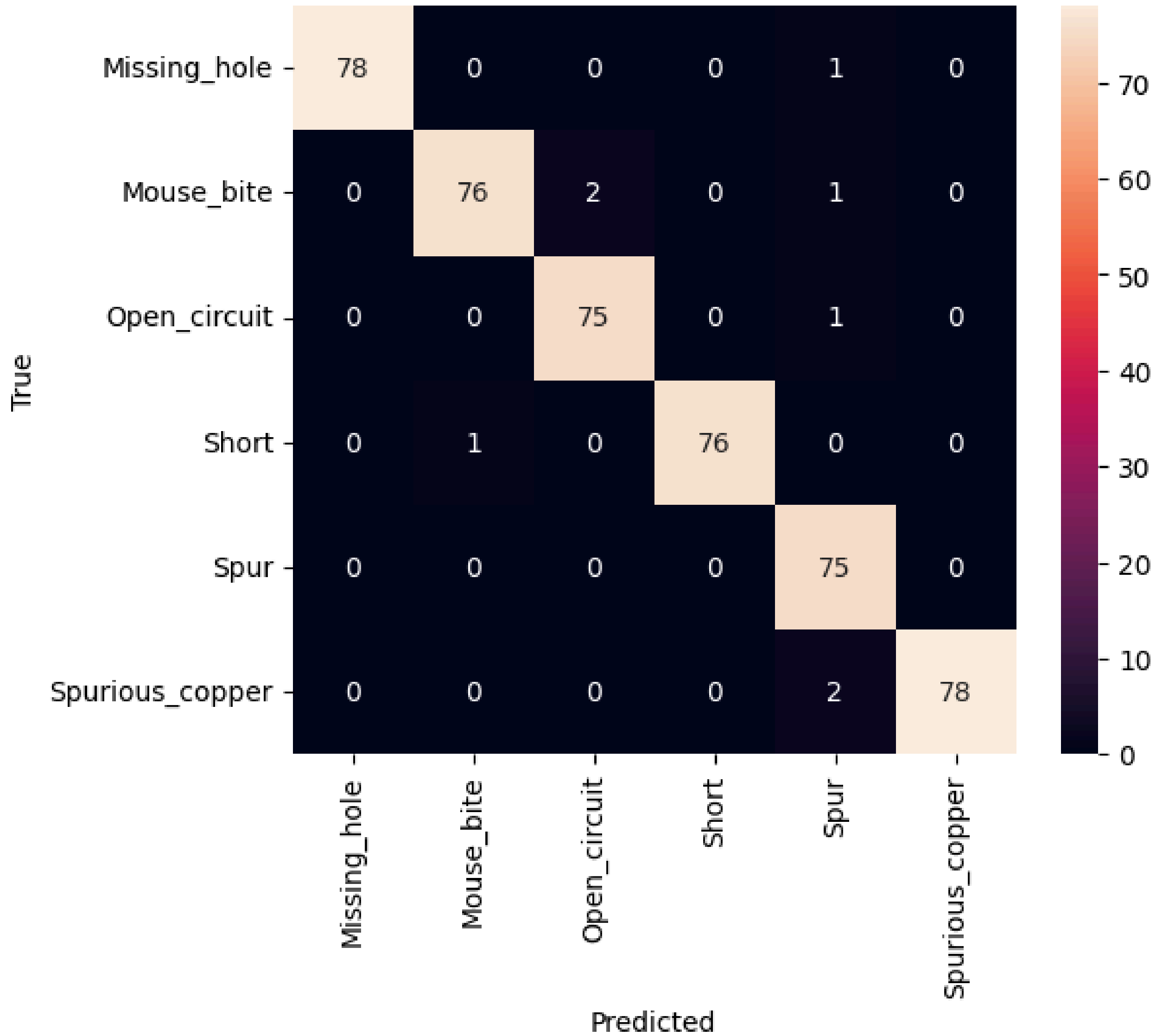


Confusion Matrix for PCB Defect Classification

As shown in the accuracy plot, both training and validation accuracy steadily increased and converged above **97%**, indicating that the model effectively learned to generalize beyond the training data. Similarly, the loss curves demonstrate a rapid decrease in both training and validation loss, stabilizing near zero confirming efficient and stable learning without signs of overfitting.

CONFUSION MATRIX ANALYSIS

The performance of the final trained model was further evaluated using a confusion matrix, which provides a detailed view of class-wise predictions:



Model Training and Validation Curves

The confusion matrix clearly shows that the majority of predictions fall along the diagonal, representing correct classifications. The model achieved **98.6% training accuracy and 97.8% validation accuracy** , with very few misclassifications observed. For example, a small number of Mouse Bite defects were predicted as *Open Circuit, and a few Spurious Copper defects were misidentified as Spur*.

These minor errors can be attributed to the visual similarity between defect types, which can sometimes make even human inspection difficult. Despite this, the model consistently demonstrated robust performance across all defect classes, maintaining a **mean per-class accuracy above 97%.**

APPLICATION TESTING AND OUTPUT VISUALIZATION

The trained model was integrated into a web-based application developed using Streamlit. Users can upload PCB images, and the system automatically detects, classifies, and annotates defect regions. The application generates real-time visual output with bounding boxes around defects and provides an option to export annotated results and prediction logs.

Testing on multiple unseen PCB samples confirmed that the system operates with an average inference time of less than 5 seconds per image, making it highly suitable for production environments.

DISCUSSION

The results demonstrate that the proposed application is highly effective in automating PCB defect detection. The system combines OpenCV-based image subtraction and contour extraction with EfficientNet deep learning classification, achieving high precision and reliability. Compared to manual inspection or costly Automated Optical Inspection (AOI) systems, this approach is:

- **Faster** in operation,
- **More cost-effective**, and
- **Easily scalable** for industrial applications.

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FUTURE ENHANCEMENT

Although the *PCB Defect Detection and Classification Application* has achieved high accuracy and efficiency, there are several potential areas for future enhancement to further improve its performance, scalability, and industrial usability.

1. Real-Time Inspection System

- The current implementation operates on static image inputs.
- Future versions can integrate real-time video inspection using industrial cameras or IoT-enabled systems to detect defects on PCBs during the production process.

2. Edge and Embedded Deployment

- The model can be optimized for lightweight deployment on edge devices such as Raspberry Pi, NVIDIA Jetson Nano, or embedded processors, enabling on-site defect detection without requiring high-end servers.

3. Extended Defect Categories

- The current system classifies six primary defects. Future datasets can include more complex and diverse defect types (e.g., component misalignment, soldering defects, bridging issues) to enhance classification coverage.

4. Automated Report Generation

- Integration of an automated reporting module that exports PDF or Excel summaries of detected defects, including classification confidence scores and timestamps, will help engineers with faster documentation and quality control.

5. Integration with Manufacturing Pipelines

- The system can be linked to automated production lines through APIs or PLC (Programmable Logic Controller) connections to immediately flag and isolate defective PCBs, reducing material wastage and downtime.

6. Model Optimization and Transfer Learning

- Further research into model compression, pruning, and quantization can reduce inference time and memory usage, allowing faster processing on low-power hardware.
- Using transfer learning with larger defect datasets can further improve model generalization across different PCB designs and manufacturers.

Enhanced User Interface

- A more interactive and responsive user interface could be developed using **ReactJS or Flask-based dashboards** , featuring real-time defect visualization, progress tracking, and analytics tools for production monitoring.

In summary, the proposed system lays a strong foundation for an intelligent, automated, and cost-effective PCB inspection framework. With continued improvements in **hardware integration, deep learning optimization, and real-time capabilities** , the project has significant potential to evolve into a full-scale **industrial-grade PCB quality assurance solution.**

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